Image Processing and Medical Imaging

Fourier transform

This work was supported by the construction EFOP-3.4.3-16-2016-00021. The project was supported by the European Union, co-financed by the European Social Fund.

UNIVERSITY of DEBRECEN

Linearity of mapping

- It is well-known from *Linear algebra,* what is linear mapping.
- Consider a mapping L between two linear spaces. It is called **additive** if $L(x+y)=Lx+Ly$ and **homogeneous** if $L(\alpha x)=\alpha L(x)$.
- We call the mapping L **linear** if it is additive and homogeneous.

UNIVERSITY of DEBRECEN

Convolution of two functions in 1D

- Convolution is an important operation in the linear approach to image analysis. The convolution is an integral which expresses the amount of overlap of one function *f(t)* as it is shifted over another function *h(t).*

$$(f * h)(t) = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau$$

UNIVERSITY of DEBRECEN

Convolution of two functions in 2D

$$(f * h)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(a, b) h(x - a, y - b)\, da\, db$$

UNIVERSITY of DEBRECEN

Properties of the convolution

$$f*h=h*f$$

$$f*(g*h)=(f*g)*h$$

$$f*(g+h)=(f*g)+(f*h)$$

$$\alpha(f*g)=(\alpha f)*g=f*(\alpha g)$$

UNIVERSITY of DEBRECEN

Linear integral transforms

- Linear integral transforms are frequently used in image processing.
- The image can be expressed as a linear combination of some basis functions of some linear integral transform.
- This is a tool which permits representations of signals and images in a more suitable domain, where information is better visible and the solution of related problems is easier.
- The most commonly used linear integral transforms in image analysis are the Fourier, cosine, and wavelet transforms.

UNIVERSITY of DEBRECEN

Fourier transform in 1D

$$F(\xi) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i \xi t} \, dt$$

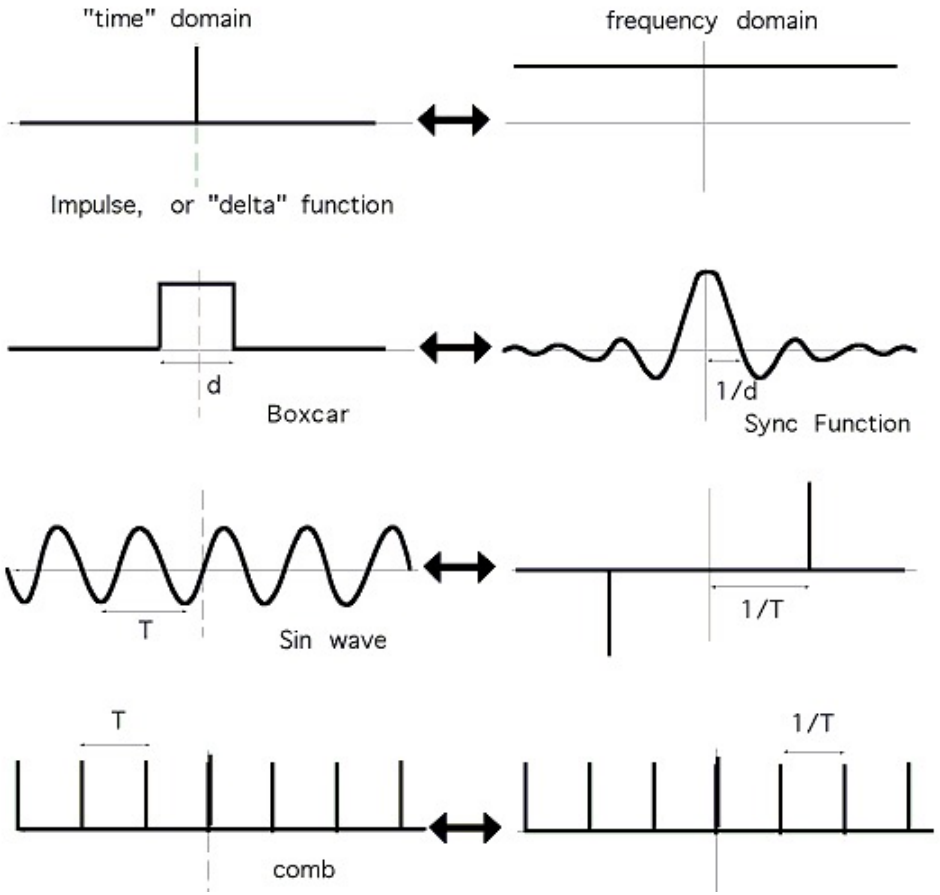
UNIVERSITY of DEBRECEN

Inverse Fourier transform in 1D

$$f(t) = \int_{-\infty}^{\infty} F(\xi) e^{2\pi i \xi t} \, d\xi$$

UNIVERSITY of DEBRECEN

Fourier transform: Examples in 1D



Source: https://www.tes.com/lessons/ubHCeL1l2MIQGA/briefs-on-data-processing

UNIVERSITY of DEBRECEN

Fourier transform and its inverse in 2D

$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)e^{-2\pi i(ux+vy)}\,dx\,dy$$

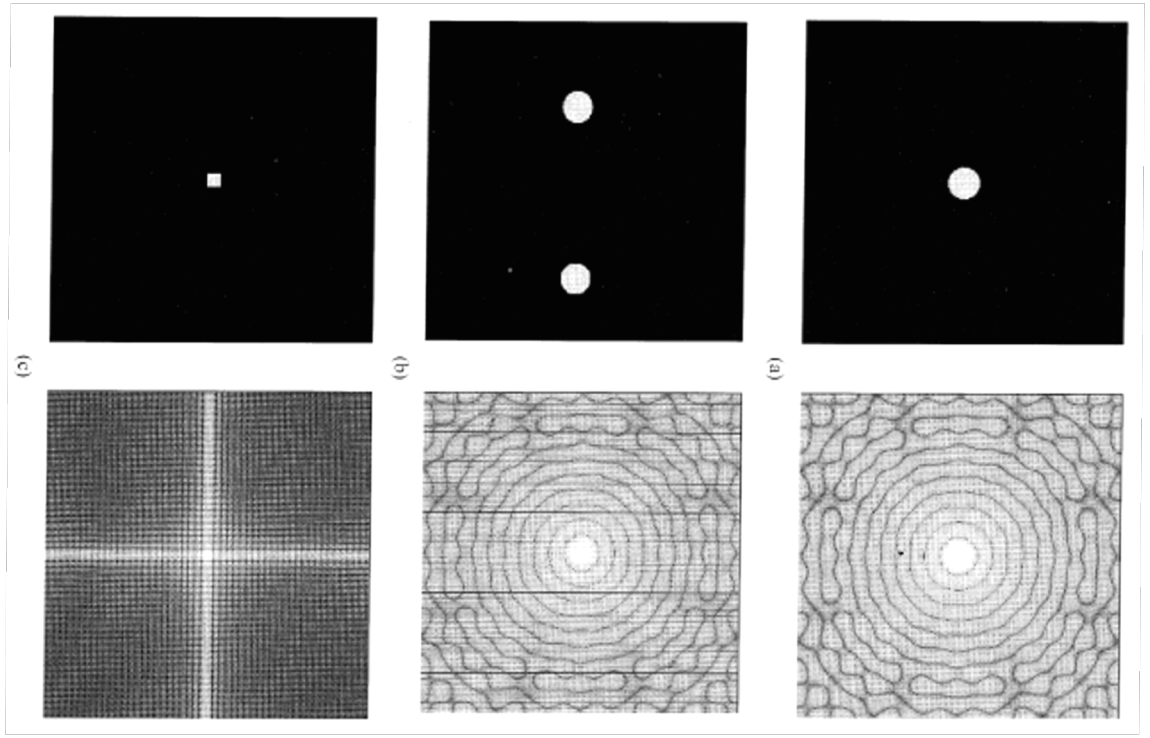
$$f(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(u,v)e^{2\pi i(ux+vy)}\,du\,dv$$

UNIVERSITY of DEBRECEN

Fourier transform: Examples in 2D

Image space

Fourier space



Source: https://math.stackexchange.com/questions/484569/how-to-interpret-the-fourier-transform-graphs-of-an-image

Some properties of the Fourier transform

- Linearity:

$$\mathscr{F}(\alpha f(x,y)+\beta g(x,y))=\alpha F(u,v)+\beta G(u,v)$$

- Shift of the origin in the image domain:

$$\mathscr{F}(f(x-a,y-b))=F(u,v)e^{-2\pi i(ua+bv)}$$

- Shift of the origin in the frequency domain:

$$\mathscr{F}(f(x,y)e^{2\pi i(ax+by)})=F(u-a,v-b)$$

UNIVERSITY of DEBRECEN

Convolution theorem

$$\mathscr{F}((f * g)(x,y))=F(u,v)G(u,v)$$

$$\mathscr{F}((f(x,y)g(x,y))=(F * G)(u,v)$$

UNIVERSITY of DEBRECEN

Discrete Fourier transform in 2D

$$F(u,v) = \frac{1}{MN}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)\exp(-2\pi i(\frac{mu}{M}+\frac{nv}{N}))$$

UNIVERSITY of DEBRECEN

Discrete inverse Fourier transform in 2D

$$f(m,n) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) \exp(2\pi i(\frac{mu}{M} + \frac{nv}{N}))$$

UNIVERSITY of DEBRECEN

Filtering in Fourier space

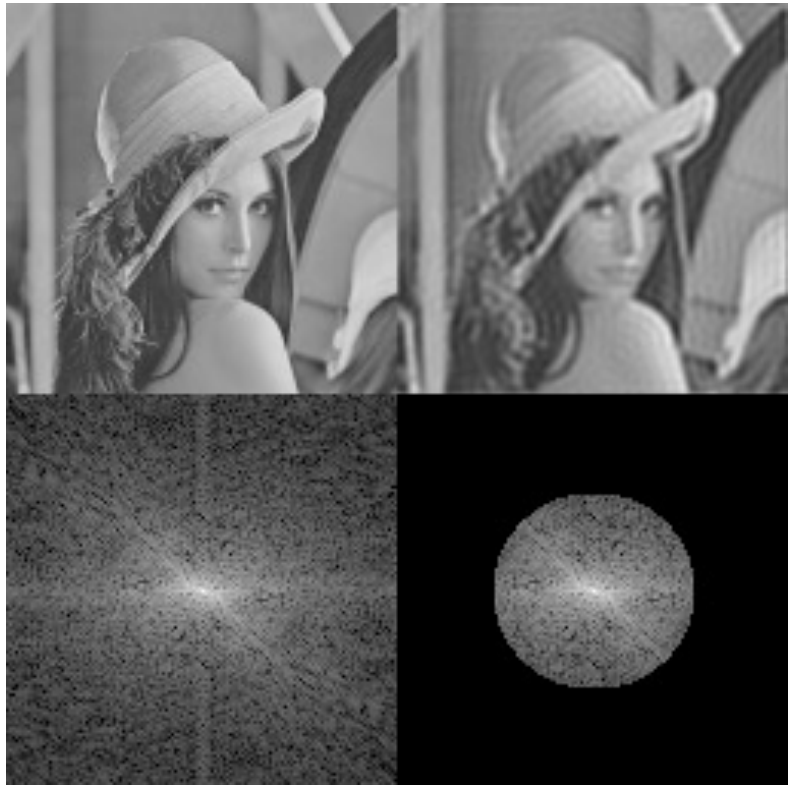
UNIVERSITY of DEBRECEN

Ideal Low Pass Filter (ILPF)

- Let us consider the following so-called filter function: $H(u,v)=1$, if $\rho(u,v)\leq\omega$, $H(u,v)=0$ otherwise.
- Here $\rho(u,v)$ is a distance function (distance between the origin and the point with coords (u,v)).
- The given value ω is the cutting frequency.

UNIVERSITY of DEBRECEN

Ideal Low Pass Filter (ILPF)



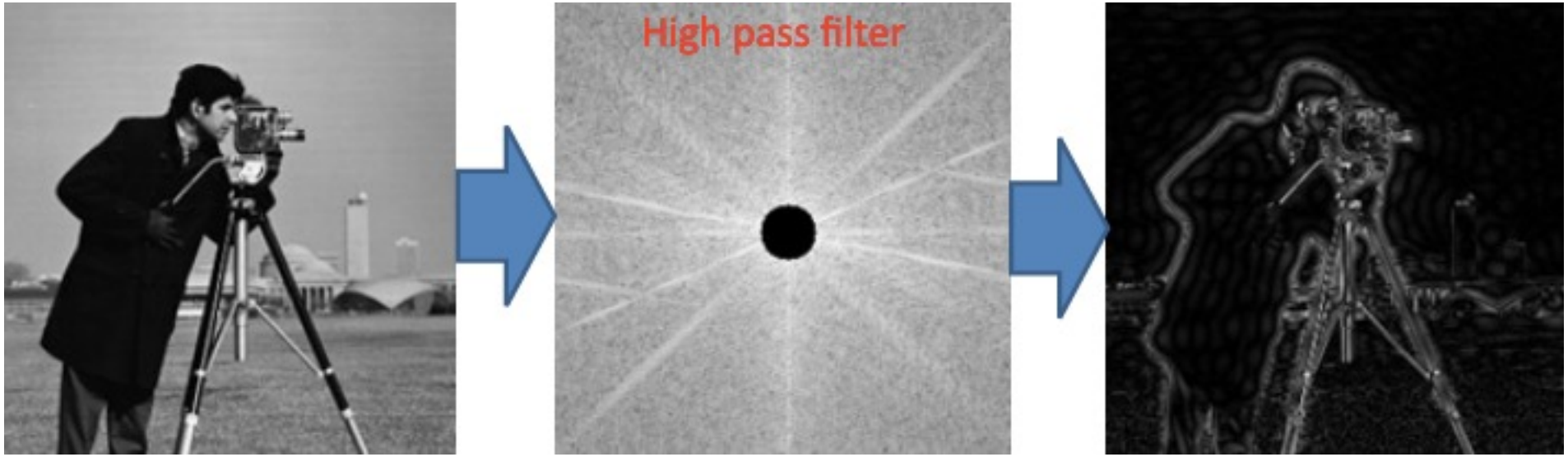
Source: http://cse19-iiith.vlabs.ac.in/theory.php?exp=fourier

Ideal High Pass Filter (IHPF)

- Let us consider the following so-called filter function: $H(u,v)=0$, if $\rho(u,v)\leq\omega$, $H(u,v)=1$ otherwise.
- Here ρ(u,v) is a distance function (distance between the origin and the point with coords (u,v)).
- The given value ω is the cutting frequency.

UNIVERSITY of DEBRECEN

Ideal High Pass Filter (IHPF)



UNIVERSITY of DEBRECEN

Source: https://slideplayer.com/slide/10660967/

Filtering in the frequency domain

- The Fourier transform is a “door” from the spatial domain into frequency domain.
- The inverse Fourier transform is a “door” from the frequency domain into spatial domain.
- We can use the filter function in the frequency domain.
- The ideal function is “not ideal”, because we are not in a continuous space.
- Band-pass filter, band-stop filter.

UNIVERSITY of DEBRECEN

How can we calculate the Fourier transform?

- Complex values, complex arithmetics.
- Round errors can produce some problems (the order of the operations is important).
- Slow running, because of time consuming calculation.
- Where is the origin? We need to shift the image, if it is necessary.
- Fast Fourier Transformation for fast calculation.

UNIVERSITY of DEBRECEN

Thank you for your attention